Practice workbook · Where a line meets a fraction curve again

Finding the other point where a tangent or normal line meets a curve whose equation is a fraction, using the point already known.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-15

1. Question 9 · part c · 4 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part c only.

9 A curve C has equation

$$y = \frac{qx - 2}{x - p} \qquad x \neq p$$

The curve crosses the y-axis at the point A.

The line l with equation $y = x + 2$ is the normal to C at A.

(a) (i) Show that $p = 1$

(ii) Find the value of q.

(7)

(b) Using the axes on the opposite page, sketch C, showing clearly the asymptotes and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(c) Find the x coordinate of D.

(4)

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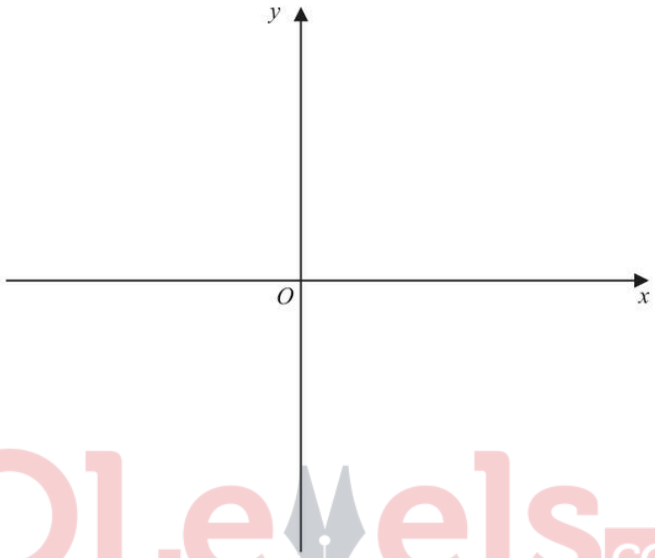
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1. Question 9 · part c · 4 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 9 continued



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1. Question 9 · part c · 4 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part c only.

Question 9 continued

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1. Question 9 · part c · 4 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part c only.



1. Question 9 · part c · 4 marks · Working and answer

Working and answer

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2. Question 7 · part c · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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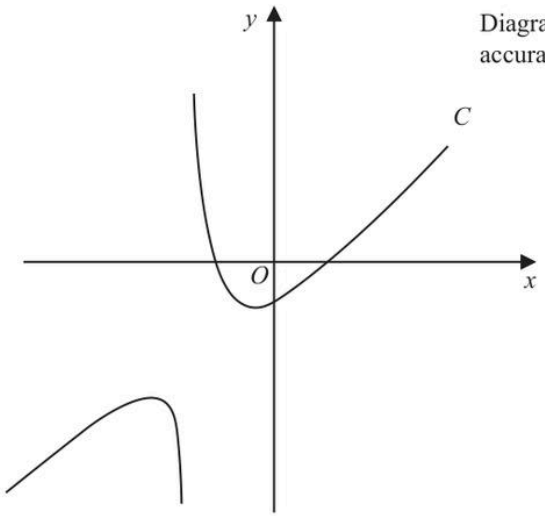


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a sketch of part of the curve C with equation

$$y = \frac{x^2 - 1}{4x + 5} \quad \text{where} \quad x \neq -\frac{5}{4}$$

(a) Write down the equation of the asymptote to C that is parallel to the y-axis.

(1)

The line l is the normal to C at the point where $x = -1$

(b) Find an equation of l

(7)

The line l meets C again at the point D

(c) Find the coordinates of D

(6)

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2. Question 7 · part c · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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2. Question 7 · part c · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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P 7 4 2 8 5 R A 0 1 6 3 2

2. Question 7 · part c · 6 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

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Question 7 continued

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(Total for Question 7 is 14 marks)

2. Question 7 · part c · 6 marks · Working and answer

Working and answer

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3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

11 A curve C has equation

$$y = \frac{(2a-1)x+1}{ax-6} \qquad \text{where } a \text{ is a constant and } x \neq \frac{6}{a}$$

(a) Find $\frac{dy}{dx}$

(3)

The curve crosses the y-axis at the point A.

The normal to C at the point A is the line l with equation $66y - 72x + 11 = 0$

Show that

(b) (i) $a = 3$

(4)

(ii) the equation of C is $y = \frac{5x+1}{3x-6}$ where $x \neq 2$

(1)

(c) Using the axes on the opposite page, sketch C, showing clearly the asymptotes with their equations and the coordinates of the points where C crosses the coordinate axes.

(5)

The line l meets C again at the point D.

(d) Find the x coordinate of D.
Give your answer as an improper fraction.

(4)

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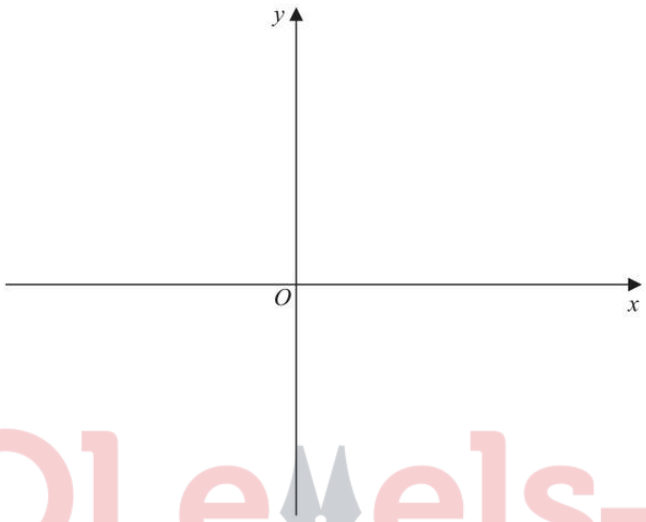
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3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued



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3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

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3. Question 11 · part d · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part d only.

Question 11 continued

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3. Question 11 · part d · 4 marks

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Further practice · Attempt part d only.

Question 11 continued

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(Total for Question 11 is 17 marks)

3. Question 11 · part d · 4 marks · Working and answer

Working and answer

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4. Question 10 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

10 The curve C with equation $y = \dfrac{6 - 3x}{x - 4}$ where $x \neq 4$, crosses the x-axis at the point P and the y-axis at the point Q

(a) Find the coordinates of

(i) P (ii) Q

(2)

(b) Write down an equation of the asymptote to C which is

(i) parallel to the y-axis (ii) parallel to the x-axis

(2)

(c) Sketch C showing clearly the asymptotes and the coordinates of the points P and Q

(3)

The line L is the normal to C at the point on C where $x = 2$

(d) Find an equation of L

(6)

The line L intersects C again at the point R

(e) Find the x coordinate of R

(3)

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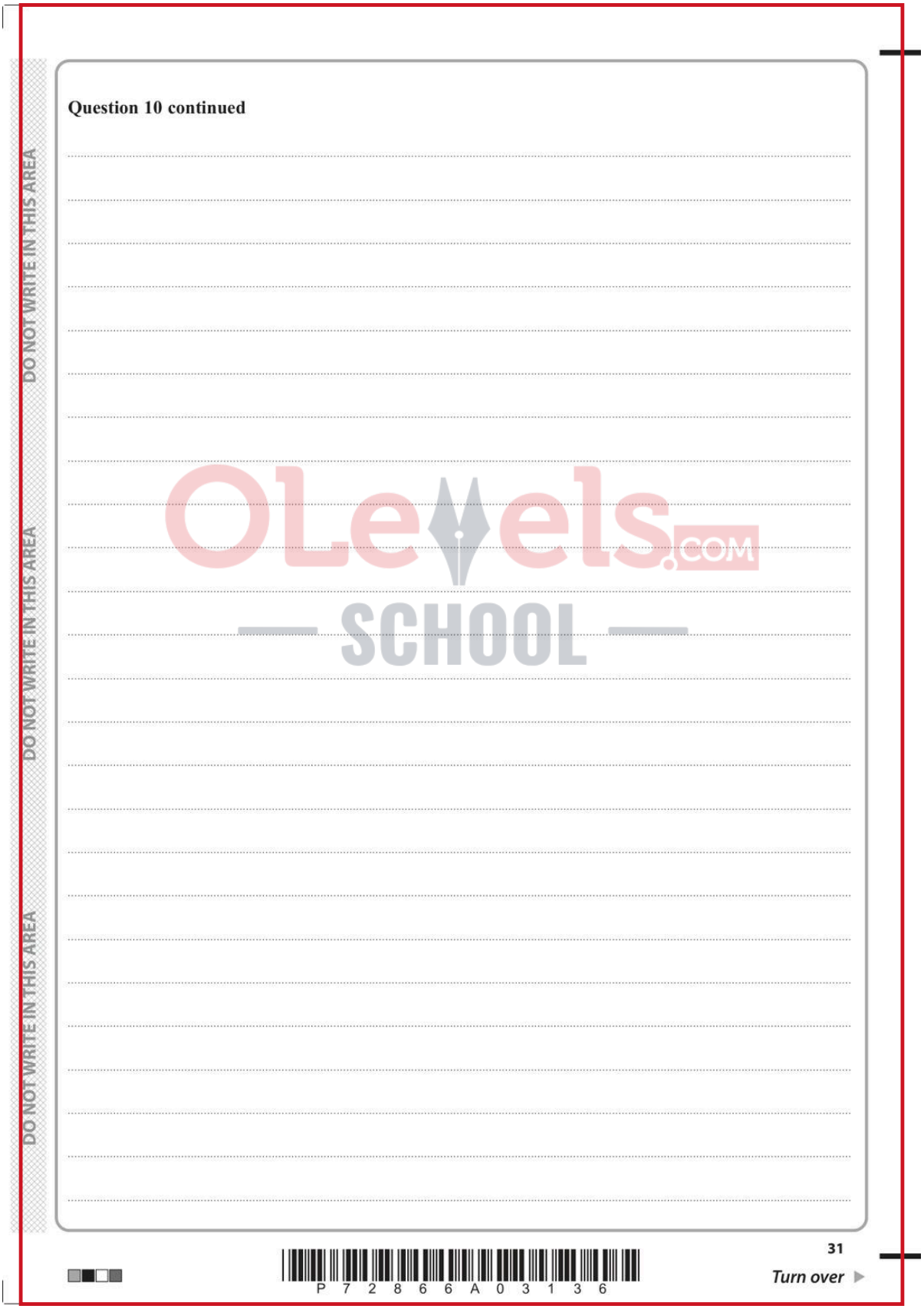
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4. Question 10 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.



4. Question 10 · part e · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part e only.

Question 10 continued

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4. Question 10 · part e · 3 marks

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Further practice · Attempt part e only.



4. Question 10 · part e · 3 marks · Working and answer

Working and answer

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